

Review of manuscript JCGS 25-127

The manuscript presents a conceptually simple yet theoretically grounded approach to addressing data heterogeneity in Federated Learning (FL). Rather than introducing a computationally heavy personalized federated learning (PFL) architecture, the authors revisit the role of within-round learning rate scheduling. By employing a decaying learning rate during local client updates, FedDecay provides a principled mechanism to flexibly balance two competing meta-learning objectives: initial shared model success (global performance) and rapid personalization (fine-tuned performance). This reframing is a strong conceptual contribution, as it offers a unifying perspective that frames standard methods like FedAvg, FedSGD, and FOMAML as special cases of a more general gradient weighting scheme, directly connecting learning rate scheduling to meta-learning theory in an interpretable manner. Further, the decomposition of the expected gradient provides the key insight for tuning the trade-off between global generalization and local adaptation, without compromise the convergence rate.

The numerical results demonstrate consistent and robust improvements in accuracy across diverse data modalities, including vision (FEMNIST), text (SST2), and graph (PUBMED) datasets.

Nevertheless, it would be helpful to examine more extreme scenarios in terms of heterogeneity; even experiments based on synthetic data would be insightful. Demonstrating that the proposed mechanism prevents model divergence in extreme corner cases would enhance the robustness claims.

Another issue that the authors should briefly investigate and comment upon is whether a client specific decay schedule would be beneficial. I understand the theoretical challenges, but I wonder if such an adaptive mechanism would could expand the personalization boundaries even further.

Finally, the manuscript would benefit from a brief discussion on whether newer FL algorithms, such as FedProx, FedAdam, and FedYogi, could also leverage this approach. Given their structural similarities to FedSGD, it seems highly likely that they would, and adding this context would be highly valuable to the readers.

Overall, the paper makes a solid contribution to a timely topic. With a few additional enhancements, I believe it will make an excellent addition to the journal.